Fisher Randomization Tests for Treatment Effects

Causal Inference Assignment II

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1 Question 1: Fisher Randomization Tests

Reanalyze the experimental data using Fisher's randomization test (FRT) with different test statistics. The null hypothesis is that the treatment has no effect on any individual:

$$H_0 : \tau_i = 0, \quad \forall i = 1, 2, \ldots, n.$$

1.1 Load and Prepare the Data

```
lalonde <- read.table(
  "C:/Users/USER/Downloads/cps1re74.csv",
  header = TRUE
) |>
  as_tibble() |>
  mutate(
    u74 = as.numeric(re74 == 0),
    u75 = as.numeric(re75 == 0)
  )
```

```r
head(lalonde)
```

```
# A tibble: 6 x 12
  treat   age  educ black hispan married nodegree  re74  re75   re78   u74   u75
  <int> <int> <int> <int>  <int>   <int>    <int> <dbl> <dbl>  <dbl> <dbl> <dbl>
1     1    37    11     1      0       1        1     0     0  9930.     1     1
2     1    22     9     0      1       0        1     0     0  3596.     1     1
3     1    30    12     1      0       0        0     0     0 24909.     1     1
4     1    27    11     1      0       0        1     0     0  7506.     1     1
5     1    33     8     1      0       0        1     0     0   290.     1     1
6     1    22     9     1      0       0        1     0     0  4056.     1     1
```

Define the number of observations and the number of treated individuals.

```r
n <- nrow(lalonde)
n1 <- sum(lalonde$treat == 1)

treat_permut <- function(n, n1) {
  A <- rep(0, n)
  A[sample(1:n, n1)] <- 1
  return(A)
}
```

1.2 Approach I: Residual-Based Test Statistic

First, regress the outcome on all available covariates, excluding the treatment variable, and obtain the residuals.

```r
model <- lm(re78 ~ . - treat, data = lalonde)
res <- model$residuals
```

Define the test statistic as the absolute difference between the mean residuals of the treatment and comparison groups:

$$T = |\overline{e}_1 - \overline{e}_0|,$$

where $\overline{e}_1$ and $\overline{e}_0$ denote the mean residuals in the treated and untreated groups, respectively.

```r
abs_res_diff <- function(A, res) {
  abs(mean(res[A == 1]) - mean(res[A == 0]))
}

I_0 <- abs_res_diff(lalonde$treat, res)

I_0
```

```
[1] 931.5196
```

Generate random treatment assignments while keeping the number of treated individuals fixed. Use these assignments to approximate the randomization distribution under the null hypothesis.

```
set.seed(007)

B <- 10000

I <- numeric(B)

for (i in 1:B) {
  comb <- treat_permut(n, n1)
  I[i] <- abs_res_diff(comb, res)
}

p_val_I <- mean(I >= I_0)

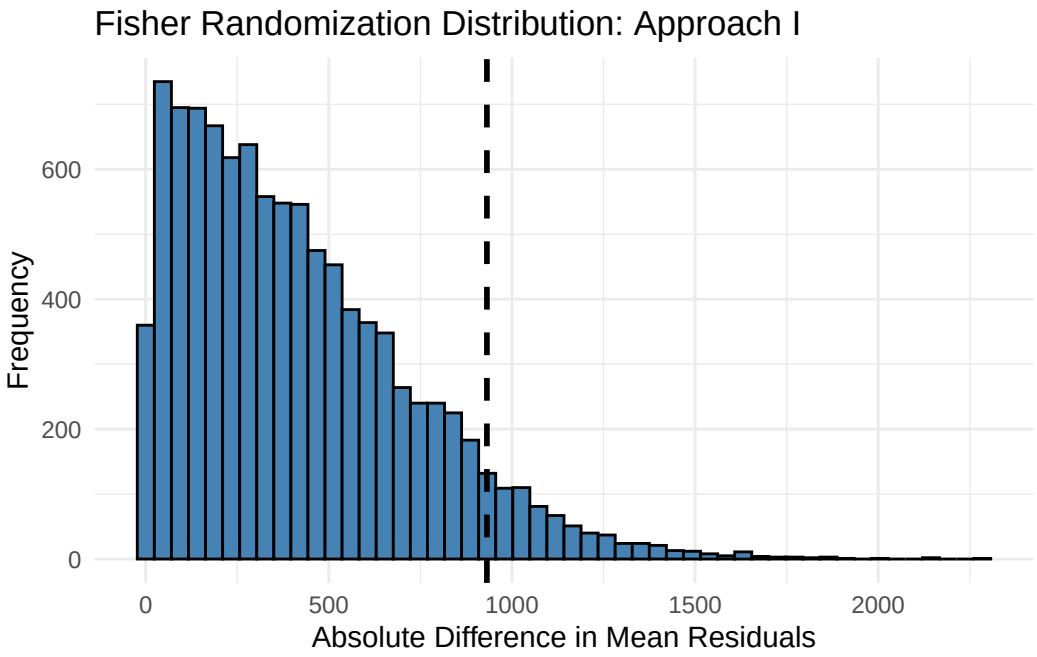
p_val_I
```

```
[1] 0.0693
```

The Fisher randomization p-value is the proportion of permuted test statistics that are at least as large as the observed statistic.

1.2.1 Randomization Distribution

```
ggplot(data.frame(I = I), aes(x = I)) +
  geom_histogram(
    bins = 50,
    color = "black",
    fill = "steelblue"
  ) +
  geom_vline(
    xintercept = I_0,
    linewidth = 1,
    linetype = "dashed"
  ) +
  labs(
    title = "Fisher Randomization Distribution: Approach I",
    x = "Absolute Difference in Mean Residuals",
    y = "Frequency"
  ) +
  theme_minimal()
```



1.3 Approach II: Treatment Coefficient as the Test Statistic

Fit a linear regression model of the outcome on treatment and all available covariates. Use the estimated treatment coefficient as the observed test statistic.

$$\text{re78}_i = \beta_0 + \beta_1 \text{treat}_i + \sum_{j=2}^{p} \beta_j X_{ij} + \varepsilon_i,$$

where β_1 is the treatment coefficient.

```
model2 <- lm(re78 ~ ., data = lalonde)

coef_0 <- coef(model2)["treat"]

coef_0
```

```
   treat
1067.546
```

Permute the treatment assignments while keeping the number of treated individuals fixed, refit the regression model, and record the treatment coefficient for each permutation.

```r
set.seed(007)

coef <- numeric(B)
data <- lalonde

for (i in 1:B) {
  data$treat <- treat_permut(n, n1)

  fit <- lm(re78 ~ ., data = data)

  coef[i] <- coef(fit)["treat"]
}

p_value_II <- mean(abs(coef) >= abs(coef_0))

p_value_II
```

```
[1] 0.0385
```

The two-sided Fisher randomization p-value is

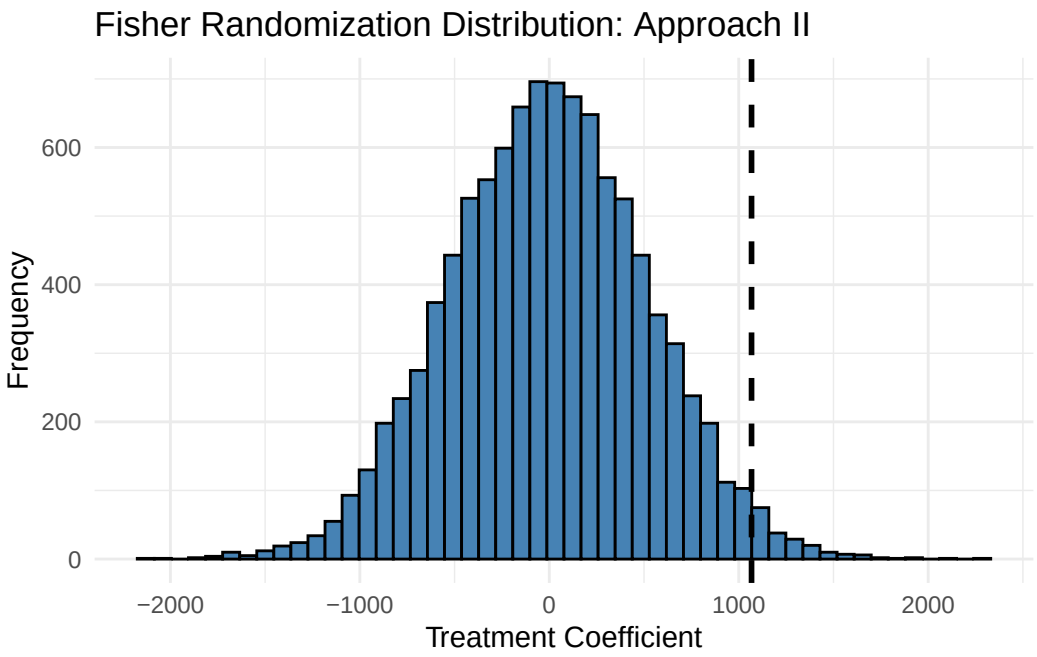
$$p = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1} \left(|T_b| \geq |T_{\text{obs}}| \right),$$

where T_{obs} is the observed treatment coefficient and T_b is the coefficient obtained from the bth permuted dataset.

1.3.1 Randomization Distribution

```r
ggplot(data.frame(coef = coef), aes(x = coef)) +
  geom_histogram(
    bins = 50,
    color = "black",
    fill = "steelblue"
  ) +
  geom_vline(
    xintercept = coef_0,
    linewidth = 1,
    linetype = "dashed"
  ) +
  labs(
    title = "Fisher Randomization Distribution: Approach II",
    x = "Treatment Coefficient",
    y = "Frequency"
```

```
  ) +
  theme_minimal()
```



1.4 Approach III: Binary Outcome and Logistic Regression

Create a binary outcome that equals 1 when `re78` is positive and 0 otherwise.

```
data <- lalonde |>
  mutate(bin78 = ifelse(re78 > 0, 1, 0))
```

Fit a logistic regression model of the binary outcome on treatment and the covariates, excluding the original continuous outcome `re78`.

The model is

$$\log\left(\frac{P(\text{bin78}_i = 1)}{1 - P(\text{bin78}_i = 1)}\right) = \beta_0 + \beta_1 \text{treat}_i + \sum_{j=2}^{p} \beta_j X_{ij}.$$

```
logit <- glm(
  bin78 ~ . - re78,
  data = data,
  family = binomial(link = "logit")
)

summary(logit)
```

```
Call:
glm(formula = bin78 ~ . - re78, family = binomial(link = "logit"),
    data = data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.734e+00  2.094e-01  13.059  < 2e-16 ***
treat        6.215e-01  2.060e-01   3.018 0.002546 **
age         -4.534e-02  2.694e-03 -16.825  < 2e-16 ***
educ        -1.018e-03  1.330e-02  -0.077 0.939014
black       -1.322e-01  9.398e-02  -1.406 0.159651
hispan      -2.041e-02  1.036e-01  -0.197 0.843788
married     -2.500e-01  6.668e-02  -3.750 0.000177 ***
nodegree     5.045e-01  8.554e-02   5.898 3.67e-09 ***
re74         3.284e-05  5.852e-06   5.612 2.00e-08 ***
re75         6.183e-05  6.024e-06  10.264  < 2e-16 ***
u74         -5.180e-01  8.420e-02  -6.152 7.64e-10 ***
u75         -1.221e+00  8.264e-02 -14.778  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 12927.5  on 16176  degrees of freedom
Residual deviance:  9951.6  on 16165  degrees of freedom
AIC: 9975.6

Number of Fisher Scoring iterations: 5
```

The coefficient of `treat` represents the estimated change in log-odds of having positive earnings in 1978, conditional on the other predictors. Its statistical significance can be assessed using the coefficient's p-value in the regression summary.

1.4.1 Fisher Randomization Test for the Logistic Regression Coefficient

Use the treatment coefficient from the logistic regression as the test statistic. Randomly permute the treatment assignments, refit the logistic regression for each permutation, and calculate the two-sided randomization p-value.

```r
set.seed(007)

B <- 2000

coef_0 <- coef(logit)["treat"]
```

```r
coef <- numeric(B)

for (i in 1:B) {
  data$treat <- treat_permut(n, n1)

  fit <- glm(
    bin78 ~ . - re78,
    data = data,
    family = binomial(link = "logit")
  )

  coef[i] <- coef(fit)["treat"]
}

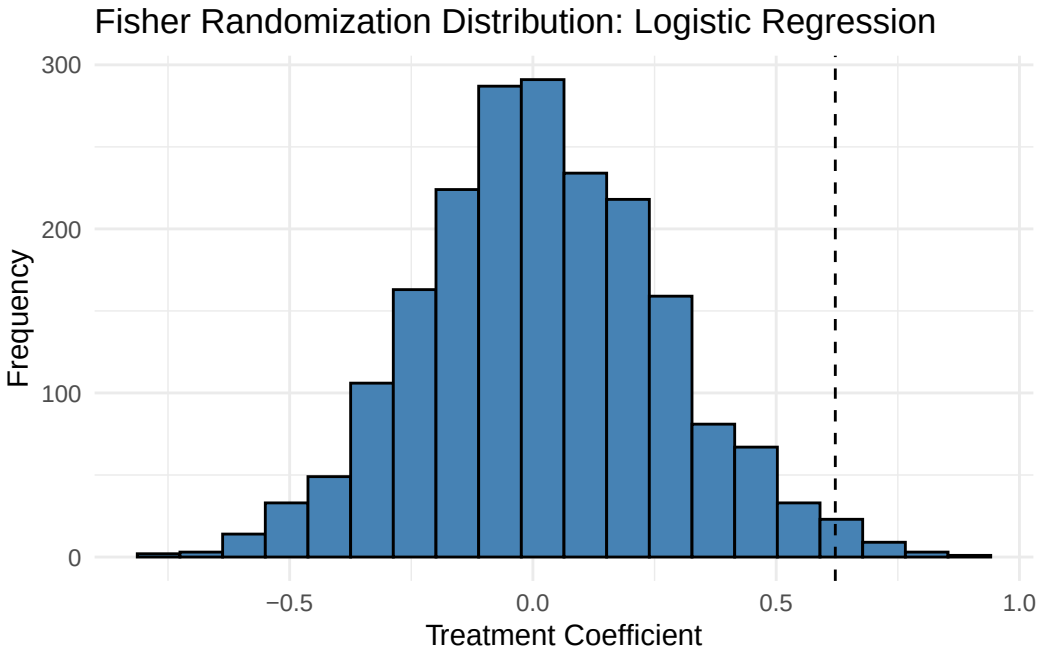
p_val_III <- mean(abs(coef) >= abs(coef_0))

p_val_III
```

```
[1] 0.0155
```

1.4.2 Randomization Distribution

```r
ggplot(data.frame(coef = coef), aes(x = coef)) +
  geom_histogram(
    bins = 20,
    fill = "steelblue",
    color = "black"
  ) +
  geom_vline(
    xintercept = coef_0,
    linetype = "dashed"
  ) +
  labs(
    title = "Fisher Randomization Distribution: Logistic Regression",
    x = "Treatment Coefficient",
    y = "Frequency"
  ) +
  theme_minimal()
```


Fisher Randomization Distribution: Logistic Regression
Frequency
300
200
100
0
−0.5
0.0
0.5
1.0
Treatment Coefficient